

Sri Lanka Institute of Information Technology



IT1040 - Fundamentals of Computing
Year 1, Semester 1- 2024

Automated Pet Feeding System APFie

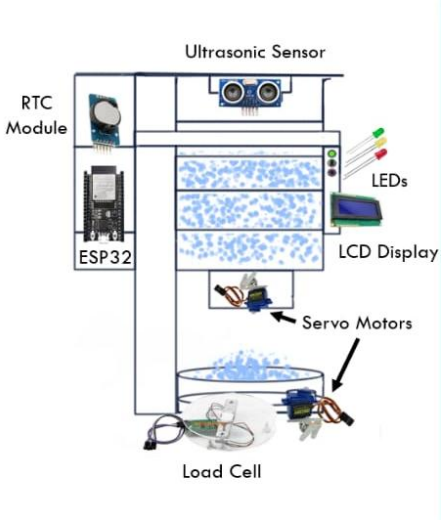
Progress Report
Group 019

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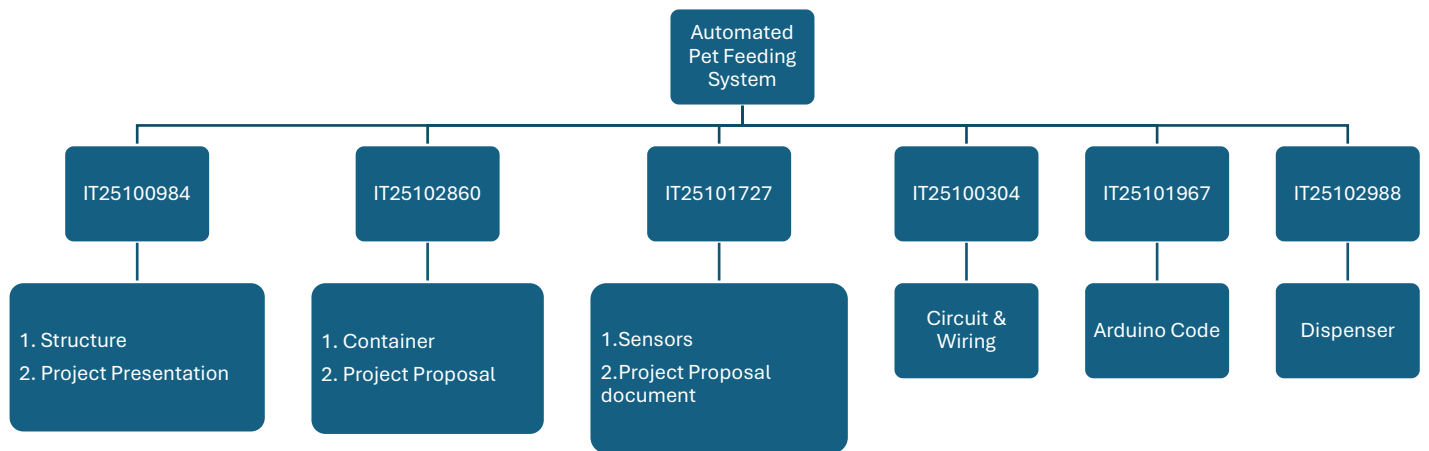
1. Introduction

The Automated Pet Feeding System is built to make feeding pets easier and more reliable, especially when their owners aren't around. Using multiple sensors, predefined weights and set times by the user, pet meal portions are dispensed accurately and efficiently. The menu display on the LCD, helps users to navigate and modify meal portions and times for their own pets. To ensure consistency of meal portions, a load cell is used in coordination with the servo motors. Along with the proximity sensor the current food level is monitored in real time. As shown in the system diagram, the RTC module keeps track of time, enabling precise meal times and offline functionality. With these features encapsulated in our device, we ensure pets are well fed, making pet care smarter and more convenient for today's busy lifestyles.

2. System Architecture

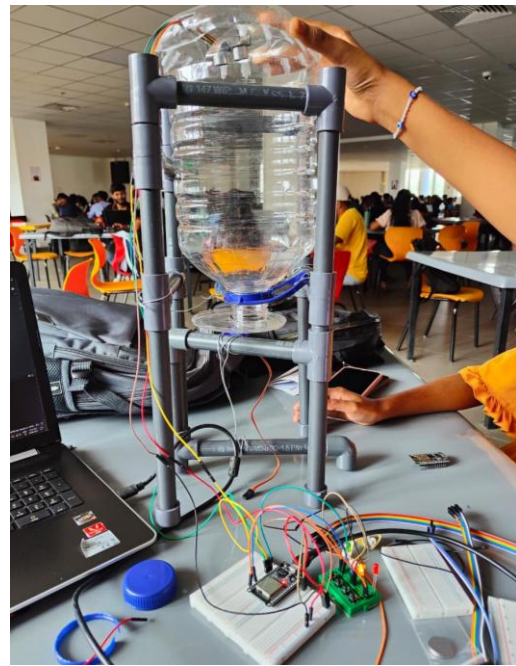
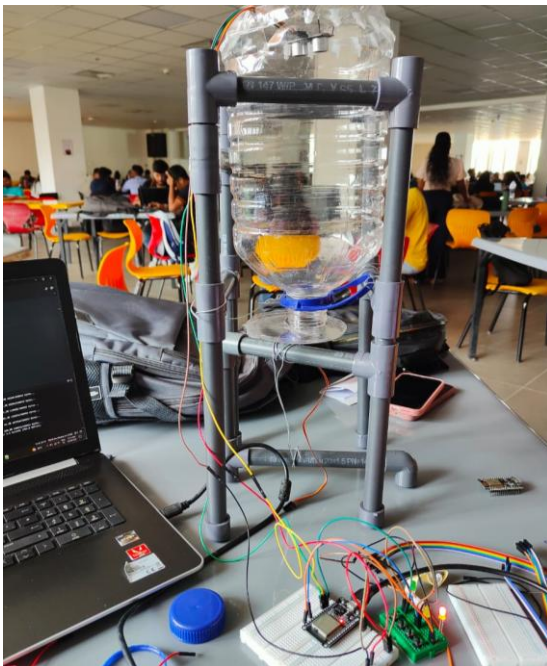
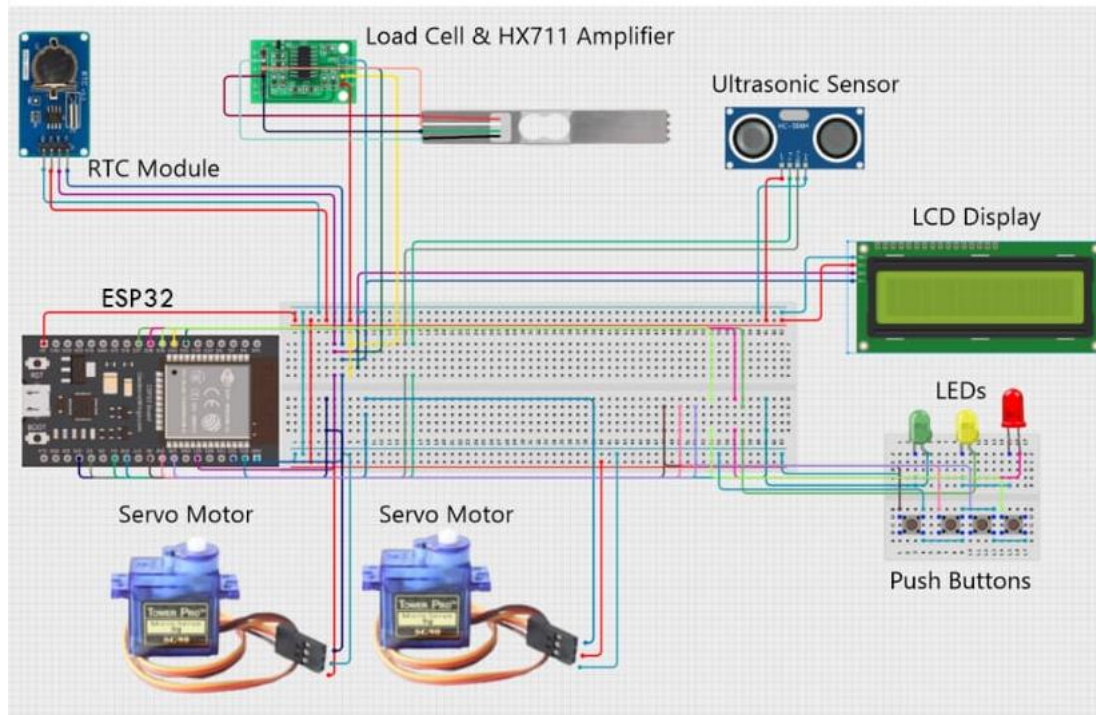
System architecture:	Description :
	With the use of multiple sensors, and predefined weights and set times by the user, pet meal portions are dispensed accordingly. The menu display on the LCD display, helps users to navigate and modify meal portions and times for their own pets. To ensure consistency of meal portions, a load cell is used in coordination with the servo motors. Along with the proximity sensor the current food level is monitored in real time. As shown in the system diagram, the RTC module keeps track of time, enabling precise meal times and offline functionality. With these features encapsulated in our device, we ensure pets are well fed.

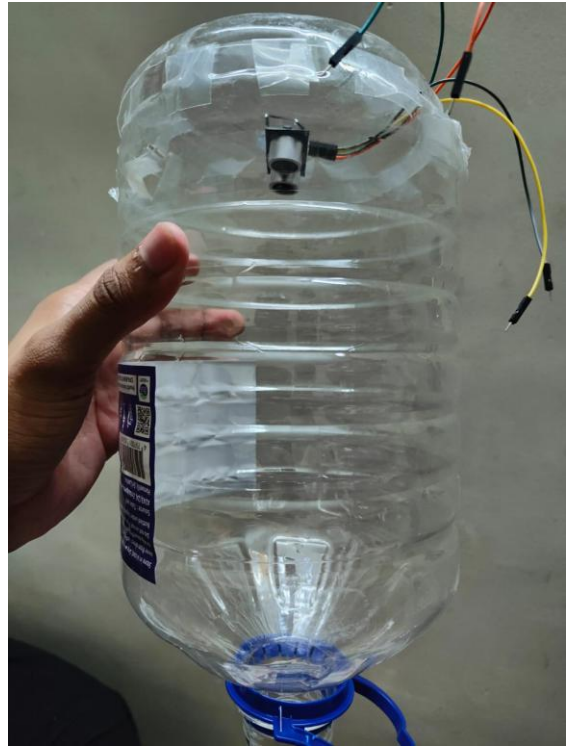
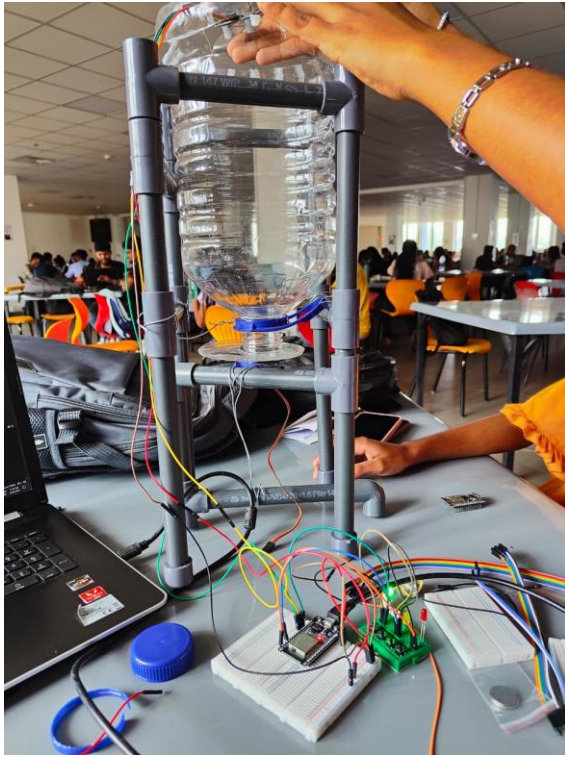
3. Work Breakdown Structure



4. Proof of Work

Circuit Diagram:





Note:

The above pictures (pic 1,2,3) illustrate the lighting of LED colors changing with regards to the proximity sensor data."